Mitsue Project — 御杖プロジェクト

A 25-Year Initiative for Forest Restoration, Distributed Renewable Energy, and Community-Owned Digital Infrastructure in Rural Japan.

Location	Mitsue Village, Nara Prefecture, Japan
Initiated	April 2026
Horizon	25 years
Current Phase	Phase 0 — Pre-Foundation (Months 1–3)
Project Lead	Rob Oudendijk (YR-Design / Safecast)
Document Status	Working draft, May 11, 2026

1. Executive Summary

The Mitsue Project is a non-profit initiative to repurpose the closed Mitsue Elementary School and the surrounding forested landscape into an integrated demonstration of rural revitalization. The project combines three mutually reinforcing activities — native forest restoration, locally generated biomass CHP energy from sugi forest thinnings (complemented by solar, EV charging, and community backup power), and a small-scale community-owned data center — under a single coordinating organization.

The project is designed to be **modest in scale, fully transparent, and openly replicable**, so that other depopulating municipalities in Japan and beyond can adapt the model to their own circumstances. The 25-year horizon is intentional: it bridges the period between today's rural energy and digital deficits and the anticipated availability of localized small-scale fusion power generation.

2. Mission and Vision

Mission. To demonstrate that rural Japanese communities can build their own sustainable future by integrating ecological restoration, locally generated clean energy, and modern digital infrastructure — and to share what is learned so that other communities may follow.

Vision (2050). Mitsue Village is a self-sustaining model of rural revitalization in which restored native forests, locally generated biomass CHP energy from sugi thinnings, complementary solar and EV charging infrastructure, community backup power, and a small-scale community-owned data center together support the village's economy, ecology, and digital future. The model is openly documented and freely available to any community that wishes to adapt it.

3. Strategic Rationale

- **Energy transition.** Within roughly ten years, the majority of Japanese passenger vehicles are expected to be electric. This transition will require significant new distributed generation capacity, particularly in rural regions where grid extension is slow and capital-intensive.
- **Forest liability conversion.** Aged sugi (cedar) plantations across rural Japan represent an under-managed asset that imposes ecological costs (pollen burden, biodiversity loss) and physical risks (landslide and fire). Active management converts liability into feedstock and timber revenue.
- **Stranded community assets.** Closed schools, such as the former Mitsue Elementary School, currently impose net maintenance costs on shrinking municipal budgets. Productive reuse turns these into community-anchored facilities.
- **Digital infrastructure deficit.** Rural broadband and edge-compute capacity continue to lag urban Japan. A small, energy-aligned data center addresses both the connectivity and the on-site computation gap.

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4. Programme Components

4.1 Forest Restoration

Phased replacement of aged sugi plantations with native broadleaf species, in cooperation with private landowners, the Forestry Agency (林野庁), and local forestry contractors. Restoration operates on a 25-year ecological timeline. Native broadleaf trees — oak, chestnut, and konara — produce acorns and nuts that sustain deer, wild boar, and bear through winter. When the forest feeds them, they stay in the forest, directly reducing wildlife raids on surrounding crop areas.

4.2 Biomass CHP Energy, Solar, EV Charging & Community Resilience

The village's primary energy source is a biomass combined heat and power (CHP) system fuelled by sugi forest thinnings, generating 24/7 baseload electricity (primary output) and heat (secondary output). This circular local energy economy — where the forest powers the village — is complemented by solar generation at the school site, EV charging stations for residents and visitors, and backup power for critical community facilities during grid outages. Unlike intermittent solar alone, biomass CHP supplies the round-the-clock baseload the data center needs. As Japan's vehicle fleet shifts to electric, local charging infrastructure becomes essential — particularly in rural areas where grid extension is slow. Battery storage will be evaluated during the Phase 1 feasibility study and installed only if confirmed.

4.3 Community-Owned Data Center

Repurposing of the closed Mitsue Elementary School building as a small-scale, energy-efficient edge-compute facility, powered entirely by locally generated renewable energy. The facility is sized for accountability and community ownership, not for hyperscale economics. By integrating a community-owned AI data center with locally generated biomass CHP and solar power, native forest restoration, and EV charging infrastructure, Mitsue sets a working example of how rural communities can combine technological progress with ecological sustainability — not as opposites, but as a single coherent system built to be replicated.

4.4 EV Charging Network

Provision of distributed charging infrastructure for residents and visitors, anchored to local generation rather than dependent on external grid extension.

4.5 Education and Open Knowledge

All methods, data, financial records, and lessons learned are published under open licences so that other communities may replicate or adapt the model.

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5. Phased Implementation

The first three years are organised into five phases, each gated by an explicit funding checkpoint. The full phase logic, including hold/descope branches and the funding-source map, is documented in `mitsue_phases_funding_flowchart.md`.

```
flowchart LR
    P0["Phase 0<br/>Pre-Foundation<br/>Months 1-3"]
    P1["Phase 1<br/>Foundation<br/>Months 4-9"]
    P2["Phase 2<br/>Pilot Design<br/>Months 10-18"]
    P3["Phase 3<br/>Pilot Build<br/>Months 19-30"]
    P4["Phase 4<br/>Operate &amp; Scale<br/>Months 31+"]

    G1{{"Gate 1<br/>¥3-8M"}}
    G2{{"Gate 2<br/>¥30-50M"}}
    G3{{"Gate 3<br/>¥120-290M"}}
    G4{{"Gate 4<br/>Revenue<br/>online?"}}

    H1["Hold &amp; Re-pitch"]
    H2["Hold / Descope"]
    H3["Stage Build"]

    P0 --> G1
    G1 -->|Pass| P1
    P1 --> G2
    G2 -->|Pass| P2
    P2 --> G3
    G3 -->|Pass| P3
    P3 --> G4
    G4 -->|Yes| P4

    G1 -->|Short| H1 -.-> G1
    G2 -->|Short| H2 -.-> G2
    G3 -->|Short| H3 -.-> G3
    G4 -->|Partial| P3

    classDef phase fill:#6796e6,stroke:#3655b5,stroke-
width:1px,color:#FFFFFF,font-weight:bold
    classDef gate  fill:#e58520,stroke:#A85C10,stroke-
width:1px,color:#FFFFFF,font-weight:bold
    classDef hold  fill:#353535,stroke:#FCEB6C,stroke-
width:1px,color:#DEDEDE,stroke-dasharray:4 3
```

```
    class P0,P1,P2,P3,P4 phase
    class G1,G2,G3,G4 gate
    class H1,H2,H3 hold
```

Phase	Months	Focus	Indicative Budget
0. Pre-Foundation	1–3	Local trust-building, founding team, draft charter	¥0–0.5M (self-funded)
1. Foundation	4–9	Legal entity, feasibility studies, professional advisors	¥3–8M
2. Pilot Design	10–18	Detailed engineering, partnerships, permits	¥15–30M
3. Pilot Build	19–30	First-stage construction, commissioning	¥120–290M
4. Operation & Scale	31+	Operations, monitoring, replication	Variable

Funding gates between phases are: **G1 ¥3–8M · G2 ¥30–50M · G3 ¥120–290M · G4 Operating revenue online**. Failure to clear a gate triggers a hold-and-re-pitch cycle rather than acceleration into an under-resourced phase.

A more detailed plan, including phase-by-phase deliverables, ROI framework, and risk register, is in `mitsue_implementation_plan.md`.

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6. Current Status — Phase 0

Period: Months 1–3 · **Budget:** Self-funded · **Posture:** No public announcements, no press, no website.

Completed

- Initial meeting with Vice Mayor of Mitsue (late 2025)
- Initial meeting with the local forestry group (early 2026)
- Drafted founding charter and detailed implementation plan (April 2026)
- Phase and funding-gate flowchart published (May 2026)
- Advisory commitments confirmed: Joi Ito and Ray Ozzie (confirmed May 5, 2026)

In Progress

- Identifying a Japanese co-founder with rural credibility (top priority)
- Scheduling a formal meeting with the Village Mayor
- Drafting bylaws for a 一般社団法人 (General Incorporated Association)
- Engaging a 行政書士 (administrative scrivener) in Nara

Next 30 Days

1. Approach candidate Japanese co-founder
2. Hold informal meeting with the Village Mayor

3. Initial consultations with one or two administrative scriveners
4. Finalise a two-page bilingual charter for distribution

A live working list is maintained in `mitsue_todo.xlsx` (PDF copies in English and Japanese available in the repository).

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7. Governance

Founding Members

- Rob Oudendijk — YR-Design / Safecast
- Japanese Co-founder — to be confirmed
- Additional founding members — to be confirmed (target: 3–5 total)

Advisory Board

- **Joi Ito** — former Director, MIT Media Lab
- **Ray Ozzie** — software pioneer; former Microsoft Chief Software Architect

Legal Structure

- **Today:** Pre-incorporation
- **First 6–9 months:** Incorporation as a 一般社団法人 (General Incorporated Association)
- **Months 18–24:** Planned conversion to NPO法人 (Specified Nonprofit Corporation) for greater grant access
- **Long-term:** Pursue 認定NPO法人 (Certified NPO) status for tax-deductible donations

Founding Documents

- `mitsue_founding_charter.md` — bilingual (EN/JP) founding charter
- `mitsue_founder_agreement_template.md` — founder alignment template

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8. Funding Strategy

The project pursues a five-layer funding stack, with each layer unlocked by the deliverables of the prior phase. This staged structure protects against early dependence on any single source.

Layer	Source	Year 1 Target	Year 3 Target
L1	Founder / private capital	¥3M	¥1M
L2	Government grants (NEDO, METI, Nara Prefecture, Mitsue village)	¥5M	¥80M
L3	Foundations (Nippon Foundation, Japan Fund for Global Environment, Toyota Foundation, others)	¥3M	¥20M
L4	Corporate partnerships (Dutch and Japanese; CSR-aligned)	¥0	¥30M

Layer	Source	Year 1 Target	Year 3 Target
L5	Operating revenue (hosting fees, FIT/FIP, EV charging fees, J-Credits)	¥0	¥3M
Total (illustrative)		**¥11M**	**¥134M**

These figures are planning targets, not commitments. Actual funding mix will depend on grant outcomes and partnership negotiations during Phases 1 and 2.

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9. Operating Principles

1. **Local first.** Every material decision begins with the wellbeing of Mitsue residents and landowners.
2. **Open and transparent.** Environmental data, financial records, and methodologies are published.
3. **Patient and long-term.** A 25-year horizon; no premature scaling.
4. **Replicable.** Documentation discipline is treated as a deliverable, not an afterthought.
5. **Modest in scale.** Small enough to remain accountable to the community.
6. **Non-partisan.** No political alignment; positions are confined to the project's mission.

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10. Repository Contents

This repository holds the working documents that govern the project's first three years. Key files:

File	Purpose
`README.md`	This document
`mitsue_founding_charter.md` / `.pdf`	Bilingual founding charter (EN/JP)
`mitsue_implementation_plan.md` / `.pdf`	Detailed five-phase implementation plan (EN)
`mitsue_implementation_plan_jp.md` / `.pdf`	Japanese translation of the implementation plan
`mitsue_phases_funding_flowchart.md` / `.pdf`	Phase spine and funding-gate diagrams
`mitsue_village_government_onepager.md` / `.pdf`	One-page brief for village government engagement
`mitsue_village_government_onepager_jp.md` / `.pdf`	Japanese version of the village government brief
`mitsue_mayor_meeting_talking_points.md` / `.pdf`	Preparation document for the formal mayor meeting (EN)

File	Purpose
`mitsue_mayor_meeting_talking_points_ja.md`	Japanese translation of the mayor meeting talking points
`mitsue_qa_briefing.md`	Bilingual Q&A briefing addressing six common questions on biomass CHP energy, EV charging, blackout resilience, data center, solar, and 25-year horizon
`mitsue_founder_agreement_template.md` / `.pdf`	Founder alignment template
`mitsue_project_founding_story.md` / `.pdf`	Narrative founding-story document (EN)
`mitsue_project_founding_story_jp.md` / `.pdf`	Japanese version of the founding story
`mitsue_stakeholders.md`	Stakeholder entity list and relationship map (EN), with Mermaid diagram
`mitsue_stakeholders_jp.md`	Japanese version of the stakeholder list and relationship map
`mitsue_stakeholder_graph.html`	Interactive stakeholder relationship graph (EN)
`mitsue_stakeholder_graph_jp.html`	Interactive stakeholder relationship graph (JP)
`mitsue_todo.xlsx` / `.pdf`	Working task list (English and Japanese PDFs)
`mitsue_finance.xlsx`	Financial planning workbook
`OPENPROJECT.md`	OpenProject project management setup, API reference, and Codeberg document index
`openproject_docker-compose.yml`	Docker Compose configuration for the local OpenProject instance
`openproject_backup.sh` / `openproject_restore.sh`	Backup and restore scripts for OpenProject data
`openproject_backup.json` / `openproject_backup.sql`	Most recent OpenProject work-package export and PostgreSQL dump
`openproject_robouden_theme.css`	Custom dark theme for the OpenProject UI (applied via Stylus extension)

Related working folders for forestry research, the closed-school site, and visual assets sit alongside this repository under the parent `Mitsue/` directory.

11. Contact

- **Project Lead:** Rob Oudendijk
- **Affiliations:** YR-Design · Safecast
- **Location:** Mitsue Village, Nara Prefecture, Japan
- **Public communications:** Not yet active; project remains in pre-foundation phase

Formal channels (website, dedicated email, NPO bank account) will be established at the start of Phase 1.

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12. Licence and Open Data

All project documentation, environmental data, and methodologies will be released under permissive open licences (Creative Commons for documents; appropriate open licences for data and code) consistent with the project's open-knowledge commitment.

Last updated: May 11, 2026 · Maintained by Rob Oudendijk